

Module 2: Sensors, Actuators and applications

IoT Sensing and Actuation: Introduction to Sensors, Sensor Characteristics, Sensorial Deviations, Sensing Types, Sensing Considerations, Actuators, Actuator Types, Actuator Characteristics

Domain Specific IoTs :Home Automation, Cities, Environment, Energy, Retail, Logistics, Agriculture, Industry, Health & Lifestyle

Textbooks:

1. Internet of Things: A Hands-On Approach, by Arshdeep Bahga & Vijay Madisetti, Universities Press, 2014

Reference books:

1. Introduction to IoT, by Sudip Misra, Anandarup Mukherjee, Arijit Roy, Cambridge University Press 2021
2. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things – David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Pearson Education, 2017

INTRODUCTION

A major chunk of IoT applications involves sensing in one form or the other. Almost all the applications in IoT be it a consumer IoT, an industrial IoT, or just plain hobby-based deployments of IoT solutions sensing forms the first step. Incidentally, actuation forms the final step in the whole operation of IoT application deployment in a majority of scenarios. The basic science of sensing and actuation is based on the process of transduction. Transduction is the process of energy conversion from one form to another. A transducer is a physical means of enabling transduction. Transducers take energy in any form (for which it is designed)— electrical, mechanical, chemical, light, sound, and others—and convert it into another, which may be electrical, mechanical, chemical, light, sound, and others. Sensors and actuators are deemed as transducers. For example, in a public announcement (PA) system, a microphone (input device) converts sound waves into electrical signals, which is amplified by an amplifier system (a process). Finally, a loudspeaker (output device) outputs this into audible sounds by converting the amplified electrical signals back into sound waves. Table 5.1 outlines the basic terminological differences between transducers, sensors, and actuators.

Table 2.1 Basic outline of the differences between transducers, sensors, and actuators

Parameters	Transducers	Sensors	Actuators
Definition	Converts energy from one form to another.	Converts various forms of energy into electrical signals.	Converts electrical signals into various forms of energy, typically mechanical energy.
Domain	Can be used to represent a sensor as well as an actuator.	It is an input transducer.	It is an output transducer.
Function	Can work as a sensor or an actuator but not simultaneously.	Used for quantifying environmental stimuli into signals.	Used for converting signals into proportional mechanical or electrical outputs.
Examples	Any sensor or actuator	Humidity sensors, Temperature sensors, Anemometers (measures flow velocity), Manometers (measures fluid pressure), Accelerometers (measures the acceleration of a body), Gas sensors (measures concentration of specific gas or gases), and others	Motors (convert electrical energy to rotary motion), Force heads (which impose a force), Pumps (which convert rotary motion of shafts into either a pressure or a fluid velocity).

SENSORS

Sensors are devices that can measure, or quantify, or respond to the ambient changes in their environment or within the intended zone of their deployment. They generate responses to external stimuli or physical phenomenon through characterization of the input functions (which are these external stimuli) and their conversion into typically electrical signals. For example, heat is converted to electrical signals in a temperature sensor, or atmospheric pressure is converted to electrical signals in a barometer. A sensor is only sensitive to the measured property (e.g., a temperature sensor only senses the ambient temperature of a room). It is insensitive to any other property besides what it is designed to detect (e.g., a temperature sensor does not bother about light or pressure while sensing the temperature). Finally, a sensor does not influence the measured property (e.g., measuring the temperature does not reduce or increase the temperature). Figure 2.1 shows the simple outline of a sensing task. Here, a temperature sensor keeps on checking an environment for changes. In the event of a fire, the temperature of the environment goes up. The temperature sensor notices this change in the temperature of the room and promptly communicates this information to a remote monitor via the processor.

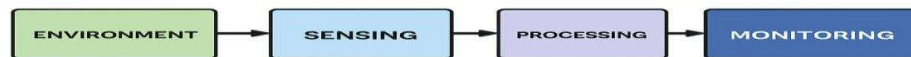


Figure 2.1 The outline of a simple sensing operation

The various sensors can be classified based on: 1) power requirements, 2) sensor output, and 3) property to be measured.

Power Requirements: The way sensors operate decides the power requirements that must be provided for an IoT implementation. Some sensors need to be provided with separate power sources for them to function, whereas some sensors do not require any power sources. Depending on the requirements of power, sensors can be of two types.

- i. **Active:** Active sensors do not require an external circuitry or mechanism to provide it with power. It directly responds to the external stimuli from its ambient environment and converts it into an output signal. For example, a photodiode converts light into electrical impulses.
- ii. **Passive:** Passive sensors require an external mechanism to power them up. The sensed properties are modulated with the sensor's inherent characteristics to generate patterns in the output of the sensor. For example, a thermistor's resistance can be detected by applying voltage

difference across it or passing a current through it.

Output: The output of a sensor helps in deciding the additional components to be integrated with an IoT node or system. Typically, almost all modern-day processors are digital; digital sensors can be directly integrated to the processors. However, the integration of analog sensors to these digital processors or IoT nodes requires additional interfacing mechanisms such as analog to digital converters (ADC), voltage level converters, and others.

Sensors are broadly divided into two types, depending on the type of output generated from these sensors, as follows.

- i. **Analog:** Analog sensors generate an output signal or voltage, which is proportional (linearly or non-linearly) to the quantity being measured and is continuous in time and amplitude. Physical quantities such as temperature, speed, pressure, displacement, strain, and others are all continuous and categorized as analog quantities. For example, a thermometer or a thermocouple can be used for measuring the temperature of a liquid (e.g., in household water heaters). These sensors continuously respond to changes in the temperature of the liquid. (ii) **Digital:** These sensors generate the output of discrete time digital representation (time, or amplitude, or both) of a quantity being measured, in the form of output signals or voltages. Typically, binary output signals in the form of logic 0 for ON or OFF, respectively are associated with digital sensors. The generated discrete (non-continuous) values may be output as a single “bit” (serial transmission), eight of which combine to produce a single “byte” output (parallel transmission) in digital sensors.
 - **Measured Property:** The property of the environment being measured by the sensors can be crucial in deciding the number of sensors in an IoT implementation. Some properties to be measured do not show high spatial variations and can be quantified only based on temporal variations in the measured property, such as ambient temperature, atmospheric pressure, and others. Whereas some properties to be measured show high spatial as well as temporal variations such as sound, image, and others. Depending on the properties to be measured, sensors can be of two types.
 - a. **Scalar:** Scalar sensors produce an output proportional to the magnitude of the quantity being measured. The output is in the form of a signal or voltage. Scalar physical quantities are those where only the magnitude of the signal is sufficient for describing or characterizing the phenomenon and information generation. Examples of such

measurable physical quantities include color, pressure, temperature, strain, and others. A thermometer or thermocouple is an example of a scalar sensor that has the ability to detect changes in ambient or object temperatures (depending on the sensor's configuration). Factors such as changes in sensor orientation or direction do not affect these sensors (typically).

- b. Vector:** Vector sensors are affected by the magnitude as well as the direction and/or orientation of the property they are measuring. Physical quantities such as velocity and images that require additional information besides their magnitude for completely categorizing a physical phenomenon are categorized as vector quantities. Measuring such quantities are undertaken using vector sensors. For example, an electronic gyroscope, which is commonly found in all modern aircraft, is used for detecting the changes in orientation of the gyroscope with respect to the Earth's orientation along all three axes.

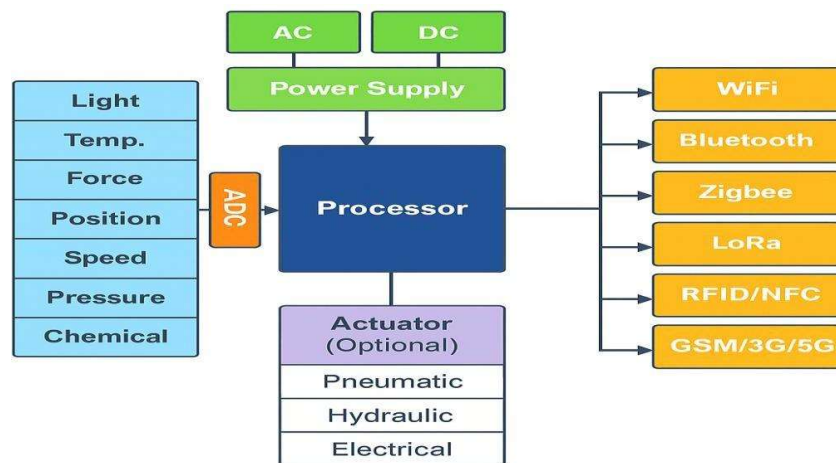


Figure 2.2 The functional blocks of a typical sensor node in IoT

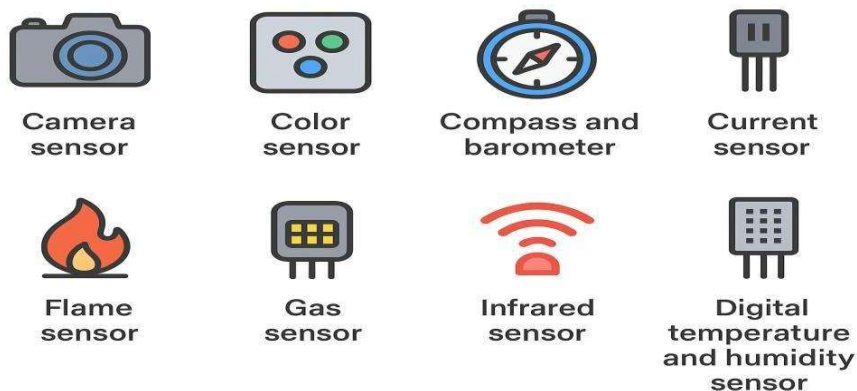


Figure 2.3 Some common commercially available sensors used for IoT-based sensing applications

SENSOR CHARACTERISTICS

All sensors can be defined by their ability to measure or capture a certain phenomenon and report them as output signals to various other systems. However, even within the same sensor type and class, sensors can be characterized by their ability to sense the phenomenon based on the following three fundamental properties.

- **Sensor Resolution:** The smallest change in the measurable quantity that a sensor can detect is referred to as the resolution of a sensor. For digital sensors, the smallest change in the digital output that the sensor is capable of quantifying is its sensor resolution. The more the resolution of a sensor, the more accurate is the precision. A sensor's accuracy does not depend upon its resolution. For example, a temperature sensor A can detect up to 0.5°C changes in temperature; whereas another sensor B can detect up to 0.25°C changes in temperature. Therefore, the resolution of sensor B is higher than the resolution of sensor A.
- **Sensor Accuracy:** The accuracy of a sensor is the ability of that sensor to measure the environment of a system as close to its true measure as possible. For example, a weight sensor detects the weight of a 100 kg mass as 99.98 kg. We can say that this sensor is 99.98% accurate, with an error rate of $\pm 0.02\%$.
- **Sensor Precision:** The principle of repeatability governs the precision of a sensor. Only if, upon multiple repetitions, the sensor is found to have the same error rate, can it be deemed as highly precise. For example, consider if the same weight sensor described earlier reports measurements of 98.28 kg, 100.34 kg, and 101.11 kg upon three repeat measurements for a mass of actual weight of 100 kg. Here, the sensor precision is not deemed high because of significant variations in the temporal measurements for the same object under the same conditions.

SENSORIAL DEVIATIONS

Most of the sensing in IoT is non-critical, where minor deviations in sensorial outputs seldom change the nature of the undertaken tasks. However, some critical applications of IoT, such as healthcare, industrial process monitoring, and others, do require sensors with high-quality measurement capabilities. As the quality of the measurement obtained from a sensor is dependent on a large number of factors, there are a few primary considerations that must be incorporated during the sensing of critical systems. In the event of a sensor's output signal going beyond its designed maximum and minimum capacity for measurement, the sensor output is truncated to its maximum or minimum value, which is also the sensor's limits. The measurement range between a sensor's

characterized minimum and maximum values is also referred to as the fullscale range of that sensor. Under real conditions, the sensitivity of a sensor may differ from the value specified for that sensor leading to sensitivity error. This deviation is mostly attributed to sensor fabrication errors and its calibration.

If the output of a sensor differs from the actual value to be measured by a constant, the sensor is said to have an offset error or bias. For example, while measuring an actual temperature of 0°C , a temperature sensor outputs 1.1°C every time. In this case, the sensor is said to have an offset error or bias of 1.1°C . Similarly, some sensors have a non-linear behavior. If a sensor's transfer function (TF) deviates from a straight-line transfer function, it is referred to as its non-linearity.

The amount a sensor's actual output differs from the ideal TF behavior over the full range of the sensor quantifies its behavior. It is denoted as the percentage of the sensor's full range. Most sensors have linear behavior. If the output signal of a sensor changes slowly and independently of the measured property, this behavior of the sensor's output is termed as drift.

Physical changes in the sensor or its material may result in long-term drift, which can span over months or years. Noise is a temporally varying random deviation of signals. In contrast, if a sensor's output varies/deviates due to deviations in the sensor's previous input values, it is referred to as hysteresis error. The present output of the sensor depends on the past input values provided to the sensor. Typically, the phenomenon of hysteresis can be observed in analog sensors, magnetic sensors, and during heating of metal strips. One way to check for hysteresis error is to check how the sensor's output changes when we first increase, then decrease the input values to the sensor over its full range. It is generally denoted as a positive and negative percentage variation of the full-range of that sensor.

Focusing on digital sensors, if the digital output of a sensor is an approximation of the measured property, it induces quantization error. This error can be defined as the difference between the actual analog signal and its closest digital approximation during the sampling stage of the analog to digital conversion. Similarly, dynamic errors caused due to mishandling of sampling frequencies can give rise to aliasing errors. Aliasing leads to different signals of varying frequencies to be represented as a single signal in case the sampling frequency is not correctly chosen, resulting in the input signal becoming a multiple of the sampling rate.

Finally, the environment itself plays a crucial role in inducing sensorial deviations. Some sensors may be prone to external influences, which may not be directly linked to the property being

measured by the sensor. This sensitivity of the sensor may lead to deviations in its output values. For example, as most sensors are semiconductor based, they are influenced by the temperature of their environment.

SENSING TYPES

Sensing can be broadly divided into four different categories based on the nature of the environment being sensed and the physical sensors being used to do so (Figure 2.4): 1) scalar sensing, 2) multimedia sensing, 3) hybrid sensing, and 4) virtual Sensing

1. Scalar sensing

Scalar sensing encompasses the sensing of features that can be quantified simply by measuring changes in the amplitude of the measured values with respect to time. Quantities such as ambient temperature, current, atmospheric pressure, rainfall, light, humidity, flux, and others are considered as scalar values as they normally do not have a directional or spatial property assigned with them. Simply measuring the changes in their values with passing time provides enough information about these quantities. The sensors used for measuring these scalar quantities are referred to as scalar sensors, and the act is known as scalar sensing. A simple scalar temperature sensing of a fire detection event is shown in Figure 2.4.

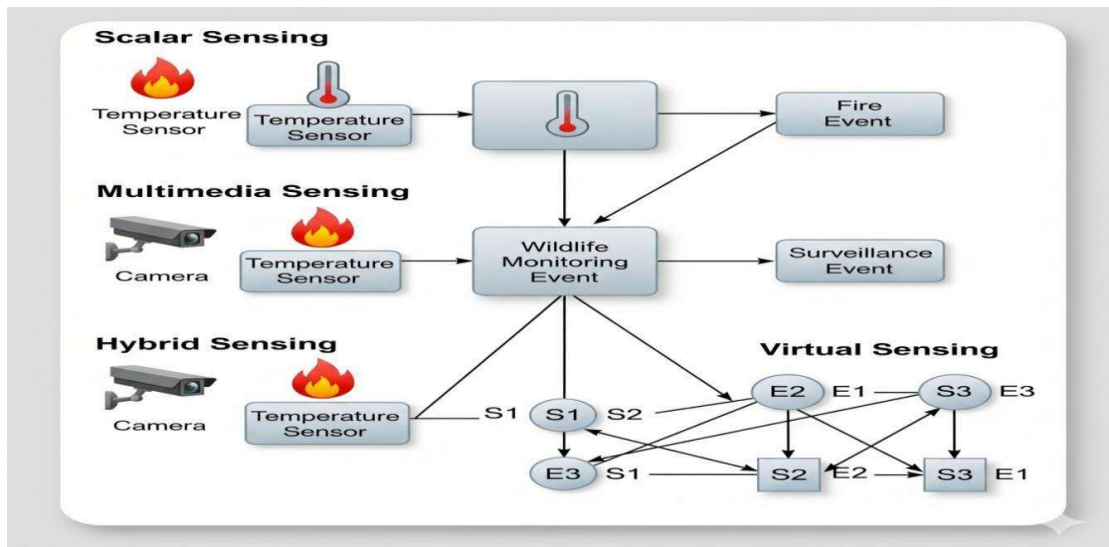


Figure 2.4 The different sensing types commonly encountered in IoT

2. Multimedia sensing

Multimedia sensing encompasses the sensing of features that have a spatial variance property associated with the property of temporal variance. Unlike scalar sensors, multimedia sensors are used for capturing the changes in amplitude of a quantifiable property concerning space (spatial) as well as time (temporal). Quantities such as images, direction, flow, speed,

acceleration, sound, force, mass, energy, and momentum have both directions as well as a magnitude. Additionally, these quantities follow the vector law of addition and hence are designated as vector quantities. They might have different values in different directions for the same working condition at the same time. The sensors used for measuring these quantities are known as vector sensors. A simple camera-based multimedia sensing using surveillance as an example is shown in Figure 2.4

3. Hybrid sensing

The act of using scalar as well as multimedia sensing at the same time is referred to as hybrid sensing. Many a time, there is a need to measure certain vector as well as scalar properties of an environment at the same time. Under these conditions, a range of various sensors are employed (from the collection of scalar as well as multimedia sensors) to measure the various properties of that environment at any instant of time, and temporally map the collected information to generate new information. For example, in an agricultural field, it is required to measure the soil conditions at regular intervals of time to determine plant health. Sensors such as soil moisture and soil temperature are deployed underground to estimate the soil's water retention capacity and the moisture being held by the soil at any instant of time. However, this setup only determines whether the plant is getting enough water or not. There may be a host of other factors besides water availability, which may affect a plant's health. The additional inclusion of a camera sensor with the plant may be able to determine the actual condition of a plant by additionally determining the color of leaves. The aggregate information from soil moisture, soil temperature, and the camera sensor will be able to collectively determine a plant's health at any instant of time. Other common examples of hybrid sensing include smart parking systems, traffic management systems, and others. Figure 2.4 shows an example of hybrid sensing, where a camera and a temperature sensor are collectively used to detect and confirm forest fires during wildlife monitoring.

4. Virtual sensing

Many a time, there is a need for very dense and large-scale deployment of sensor nodes spread over a large area for monitoring of parameters. One such domain is agriculture. Here, often, the parameters being measured, such as soil moisture, soil temperature, and water level, do not show significant spatial variations. Hence, if sensors are deployed in the fields of farmer A, it is highly likely that the measurements from his sensors will be able to provide almost concise measurements of his neighbour B's fields; this is especially true of fields which are immediately surrounding A's fields. Exploiting this property, if the data from A's field is digitized using an IoT infrastructure and this system advises him regarding the appropriate watering, fertilizer, and pesticide regimen for his crops, this advisory can also be used by B for maintaining his crops. In short, A's sensors are being used for actual measurement of parameters; whereas virtual data (which does not have actual physical sensors but uses extrapolation-based measurements) is being used for advising B. This is the virtual sensing paradigm. Figure 2.4(d) shows an example of virtual sensing. Two temperature sensors S1 and S3 monitor three nearby events E1, E2, and E3 (fires). The event E2 does not have a dedicated sensor for monitoring it; however, through the superposition of readings from sensors S1 and S3, the presence of fire in E2 is inferred.

SENSING CONSIDERATIONS

The choice of sensors in an IoT sensor node is critical and can either make or break the feasibility of an IoT deployment. The following major factors influence the choice of sensors in IoT-based sensing solutions: 1) sensing range, 2) accuracy and precision, 3) energy, and 4) device size. These factors are discussed as follows:

- **Sensing Range:** The sensing range of a sensor node defines the detection fidelity of that node. Typical approaches to optimize the sensing range in deployments include fixed k-coverage and dynamic k-coverage. A lifelong fixed k-coverage tends to usher in redundancy as it requires a large number of sensor nodes, the sensing range of some of which may also overlap. In contrast, dynamic kcoverage incorporates mobile sensor nodes post detection of an event, which, however, is a costly solution and may not be deployable in all operational areas and terrains. Additionally, the sensing range of a sensor may also be used to signify the upper and lower bounds of a sensor's measurement range. For example, a proximity sensor has a typical sensing range of a couple of meters. In contrast, a camera has a sensing range varying between tens of meters to hundreds of meters. As the complexity of the sensor and its sensing range goes up, its cost significantly increases.

- **Accuracy and Precision:** The accuracy and precision of measurements provided by a sensor are critical in deciding the operations of specific functional processes. Typically, off-the-shelf consumer sensors are low on requirements and often very cheap. However, their performance is limited to regular application domains. For example, a standard temperature sensor can be easily integrated with conventional components for hobby projects and day-to-day applications, but it is not suitable for industrial processes. Regular temperature sensors have a very low-temperature sensing range, as well as relatively low accuracy and precision. The use of these sensors in industrial applications, where a precision of up to 3– 4 decimal places is required, cannot be facilitated by these sensors. Industrial sensors are typically very sophisticated, and as a result, very costly. However, these industrial sensors have very high accuracy and precision score, even under harsh operating conditions.
- **Energy:** The energy consumed by a sensing solution is crucial to determine the lifetime of that solution and the estimated cost of its deployment. If the sensor or the sensor node is so energy inefficient that it requires replenishment of its energy sources quite frequently, the effort in maintaining the solution and its cost goes up; whereas its deployment feasibility goes down. Consider a scenario where sensor nodes are deployed on the top of glaciers. Once deployed, access to these nodes is not possible. If the energy requirements of the sensor nodes are too high, such a deployment will not last long, and the solution will be highly infeasible as charging or changing of the energy sources of these sensor nodes is not an option.
- **Device Size:** Modern-day IoT applications have a wide penetration in all domains of life. Most of the applications of IoT require sensing solutions which are so small that they do not hinder any of the regular activities that were possible before the sensor node deployment was carried out. Larger the size of a sensor node, larger is the obstruction caused by it, higher is the cost and energy requirements, and lesser is its demand for the bulk of the IoT applications. Consider a simple human activity detector. If the detection unit is too large to be carried or too bulky to cause hindrance to regular normal movements, the demand for this solution would be low. It is because of this that the onset of wearables took off so strongly. The wearable sensors are highly energy efficient, small in size, and almost part of the wearer's regular wardrobe.

ACTUATORS

An actuator can be considered as a machine or system's component that can affect the movement or control the said mechanism or the system. Control systems affect changes to the environment or property they are controlling through actuators. The system activates the actuator through a control signal, which may be digital or analog. It elicits a response from the actuator, which is in the form of some form of mechanical motion. The control system of an actuator can be a mechanical or electronic system, a software-based system (e.g., an autonomous car control system), a human, or any other input. Figure 2.5 shows the outline of a simple actuation system. A remote user sends commands to a processor. The processor instructs a motor controlled robotic arm to perform the commanded tasks accordingly. The processor is primarily responsible for converting the human commands into sequential machine-language command sequences, which enables the robot to move. The robotic arm finally moves the designated boxes, which was its assigned task.

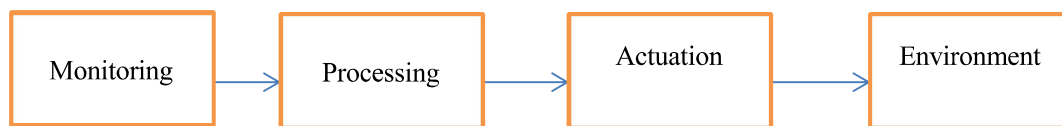


Figure 2.5 The outline of a simple actuation mechanism

ACTUATOR TYPES

Broadly, actuators can be divided into seven classes: 1) Hydraulic, 2) pneumatic, 3) electrical, 4) thermal/magnetic, 5) mechanical, 6) soft, and 7) shape memory polymers. Figure 2.6 shows some of the commonly used actuators in IoT applications.

1. Hydraulic actuators

A hydraulic actuator works on the principle of compression and decompression of fluids. These actuators facilitate mechanical tasks such as lifting loads through the use of hydraulic power derived from fluids in cylinders or fluid motors. The mechanical motion applied to a hydraulic actuator is converted to either linear, rotary, or oscillatory motion. The almost incompressible property of liquids is used in hydraulic actuators for exerting significant force. These hydraulic actuators are also considered as stiff systems. The actuator's limited acceleration restricts its usage.

2. Pneumatic actuators

A pneumatic actuator works on the principle of compression and decompression of gases. These actuators use a vacuum or compressed air at high pressure and convert it into either linear or rotary motion. Pneumatic rack and pinion actuators are commonly used for valve

controls of water pipes. Pneumatic actuators are considered as compliant systems. The actuators using pneumatic energy for their operation are typically characterized by the quick response to starting and stopping signals. Small pressure changes can be used for generating large forces through these actuators. Pneumatic brakes are an example of this type of actuator which is so responsive that they can convert small pressure changes applied by drives to generate the massive force required to stop or slow down a moving vehicle. Pneumatic actuators are responsible for converting pressure into force. The power source in the pneumatic actuator does not need to be stored in reserve for its operation.

3. Electric actuators

Typically, electric motors are used to power an electric actuator by generating mechanical torque. This generated torque is translated into the motion of a motor's shaft or for switching (as in relays). For example, actuating equipments such as solenoid valves control the flow of water in pipes in response to electrical signals. This class of actuators is considered one of the cheapest, cleanest and speedy actuator types available.

4. Thermal or magnetic actuators

The use of thermal or magnetic energy is used for powering this class of actuators. These actuators have a very high power density and are typically compact, lightweight, and economical. One classic example of thermal actuators is shape memory materials (SMMs) such as shape memory alloys (SMAs). These actuators do not require electricity for actuation. They are not affected by vibration and can work with liquid or gases. Magnetic shape memory alloys (MSMAs) are a type of magnetic actuators.

5. Mechanical actuators

In mechanical actuation, the rotary motion of the actuator is converted into linear motion to execute some movement. The use of gears, rails, pulleys, chains, and other devices are necessary for these actuators to operate. These actuators can be easily used in conjunction with pneumatic, hydraulic, or electrical actuators. They can also work in a standalone mode. The best example of a mechanical actuator is a rack and pinion mechanism.

6. Soft actuators

Soft actuators (e.g., polymer-based) consists of elastomeric polymers that are used as embedded fixtures in flexible materials such as cloth, paper, fiber, particles, and others. The conversion of molecular level microscopic changes into tangible macroscopic deformations is the primary working principle of this class of actuators. These actuators have a high stake in modern-day robotics. They are designed to handle fragile objects such as agricultural fruit harvesting, or performing precise operations like manipulating the internal organs during robot

assisted surgeries.

7. Shape memory polymers

Shape memory polymers (SMP) are considered as smart materials that respond to some external stimulus by changing their shape, and then revert to their original shape once the affecting stimulus is removed. Features such as high strain recovery, biocompatibility, low density, and biodegradability characterize these materials. SMP- based actuators function similar to our muscles. Modern-day SMPs have been designed to respond to a wide range of stimuli such as pH changes, heat differentials, light intensity, and frequency changes, magnetic changes, and others. Photopolymer/light- activated polymers (LAP) are a particular type of SMP, which require light as a stimulus to operate. LAP-based actuators are characterized by their rapid response times. Using only the variation of light frequency or its intensity, LAPs can be controlled remotely without any physical contact. The development of LAPs whose shape can be changed by the application of a specific frequency of light have been reported. The polymer retains its shape after removal of the activating light. In order to change the polymer back to its original shape, a light stimulus of a different frequency has to be applied to the polymer.

ACTUATOR CHARACTERISTICS

The choice or selection of actuators is crucial in an IoT deployment, where a control mechanism is required after sensing and processing of the information obtained from the sensed environment. Actuators perform the physically heavier tasks in an IoT deployment; tasks which require moving or changing the orientation of physical objects, changing the state of objects, and other such activities. The correct choice of actuators is necessary for the long-term sustenance and continuity of operations, as well as for increasing the lifetime of the actuators themselves. A set of four characteristics can define all actuators:

- **Weight:** The physical weight of actuators limits its application scope. For example, the use of heavier actuators is generally preferred for industrial applications and applications requiring no mobility of the IoT deployment. In contrast, lightweight actuators typically find common usage in portable systems in vehicles, drones, and home IoT applications. It is to be noted that this is not always true. Heavier actuators also have selective usage in mobile systems, for example, landing gears and engine motors in aircraft.
- **Power Rating:** This helps in deciding the nature of the application with which an actuator can be associated. The power rating defines the minimum and maximum operating power an actuator can safely withstand without damage to itself. Generally, it is indicated as the power-to-weight ratio for actuators. For example, smaller servo motors used in hobby projects typically have a

maximum rating of 5 VDC, 500 mA, which is suitable for an operations-driven battery-based power source. Exceeding this limit might be detrimental to the performance of the actuator and may cause burnout of the motor. In contrast to this, servo motors in larger applications have a rating of 460 VAC, 2.5 A, which requires standalone power supply systems for operations. It is to be noted that actuators with still higher ratings are available and vary according to application requirements.

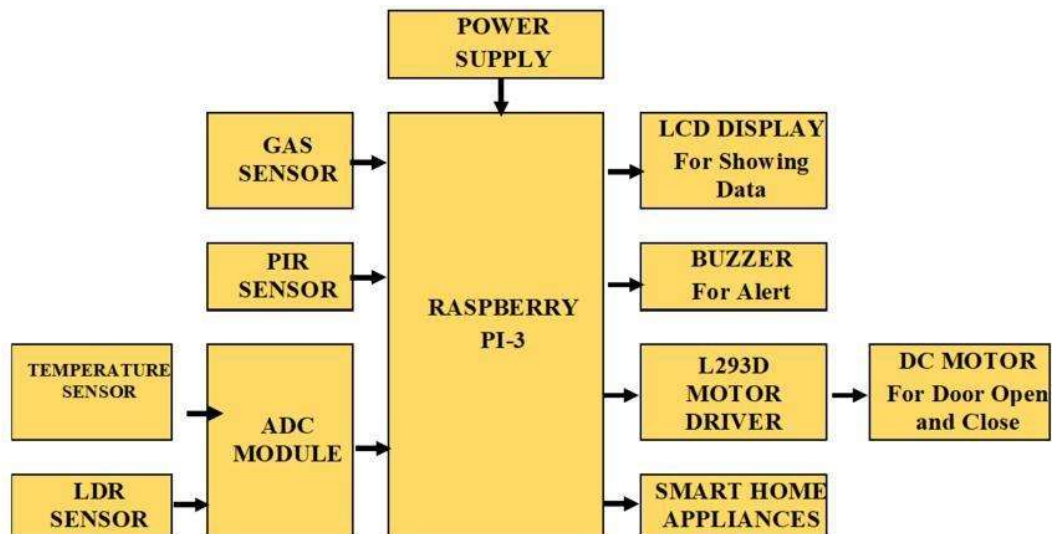
- **Torque to Weight Ratio:** The ratio of torque to the weight of the moving part of an instrument/device is referred to as its torque/weight ratio. This indicates the sensitivity of the actuator. Higher is the weight of the moving part; lower will be its torque to weight ratio for a given power.
- **Stiffness and Compliance:** The resistance of a material against deformation is known as its stiffness, whereas compliance of a material is the opposite of stiffness. Stiffness can be directly related to the modulus of elasticity of that material. Stiff systems are considered more accurate than compliant systems as they have a faster response to the change in load applied to it. For example, hydraulic systems are considered as stiff and non-compliant, whereas pneumatic systems are considered as compliant.

Introduction of Domain Specific IoTs

The Internet of Things (IoT) applications span a wide range of domains Home Automation Cities Environment, Energy, Retail, Logistics, Agriculture, Industry and Health & Lifestyle.



1. HOME AUTOMATION



1.1 Smart Lighting

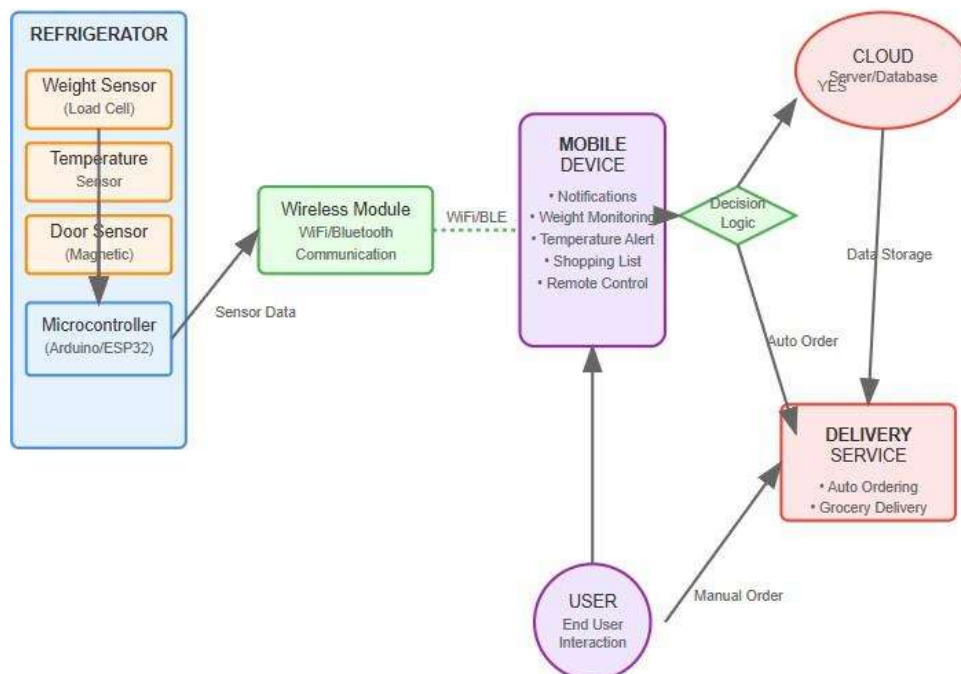
It helps in saving energy by adapting the lighting to the ambient conditions and switching on/off or dimming the light when needed. Key enabling technologies for smart lighting include solid state lighting (such as LEDs and IP enabled lights). Smart lighting solutions for home achieve energy savings by sensing the human movements and their environments and controlling the lights accordingly. Wireless enabled and Internet connected lights can be controlled remotely from IoT applications such as mobile or web applications.

1.2 Smart Appliances

Make the management easier and also provide status information to the users remotely.

- Smart appliances make the management easier and provide status information of appliances to the users remotely. **E.g:** smart washer/dryer that can be controlled remotely and notify when the washing/drying cycle is complete.
- **OpenRemote** is an open source automation platform for smart home and building that can control various appliances using mobile and web applications.
- It comprises of three components:
 - a Controller → manages scheduling and runtime integration between devices.
 - a Designer → allows to create both configuration for the controller and user interface designs.
 - Control Panel → allows to interact with devices and control them.

Smart Refrigerator IoT System - Block Diagram



1.3 Intrusion Detection

It uses security cameras and sensors (PIR sensors and door sensors) to detect intrusion and raise alerts. Alerts can be in the form of SMS or email sent to the user.

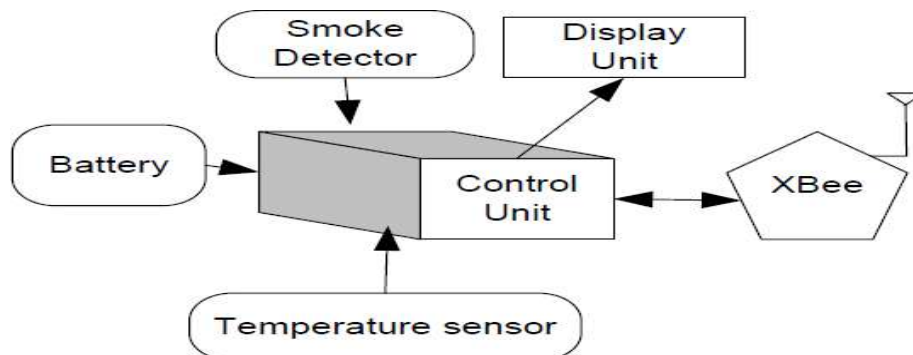
- Home intrusion detection systems use *security cameras* and *sensors* to detect intrusions and raise alerts.
- The form of the alerts can be in form:
 - *SMS*
 - *Email*
 - *Image grab or a short video clip as an email attachment*

1.4 Home Network Routers:

All the Internet connected devices in homes these days, the network router continues to be by far the most targeted in attacks. "Most Internet routers that are keystone to our home network are riddled with security issues, which make them easy picking for hackers. Most routers worldwide had default or basic username and password combinations, like "admin" and "password" while others have, they user address, birthday or name as password. Because of this, most routers are vulnerable to simple password attacks, which is basically an open invitation to malicious hackers.

1.5 Smoke/ Gas Detectors

Smoke detectors are installed in homes and buildings to detect smoke that is typically an early sign of fire. Alerts raised by smoke detectors can be in the form of signals to a fire alarm system. Gas detectors can detect the presence of harmful gases such as CO, LPG etc., Smoke detector detects the smoke in the home and it sends signal to the control unit that displays the smoke unit in display section and temperature sensor mentioned above gives the temperature of the room and signals are send through IoT to main controller just to control some activities of room.

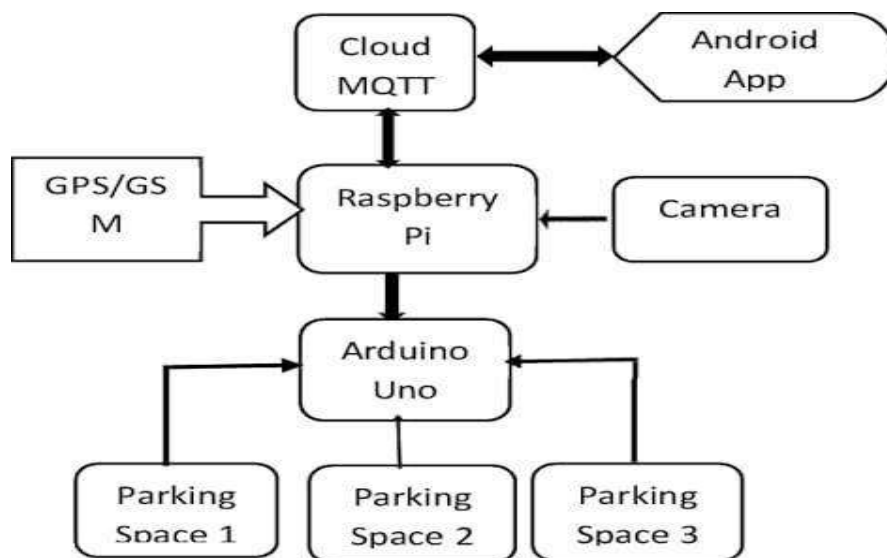


2. CITIES

2.1 Smart Parking

To make the search for parking space easier and convenient for drivers. Smart parking are powered by IoT systems that detect the no. of empty parking slots and send information over internet to smart application backends. Smart parking management system can be used to find the vacant location for a vehicle at different public places.

Smart Parking's In-Ground Vehicle Detection Sensors are core technologies, playing a key part in the Smart Parking solution that is revolutionizing how drivers in the malls and city centers can find an available parking space. Wireless sensors are embedded into parking spaces, transmitting data on the timing and duration of the space used via local signal processors into a central parking management application. Smart Parking reduces congestion, decreases vehicle emissions, lowers enforcement costs and cuts driver stress. For effective deployment of smart parking technologies, each device needs to have a reliable connectivity with the cloud servers.

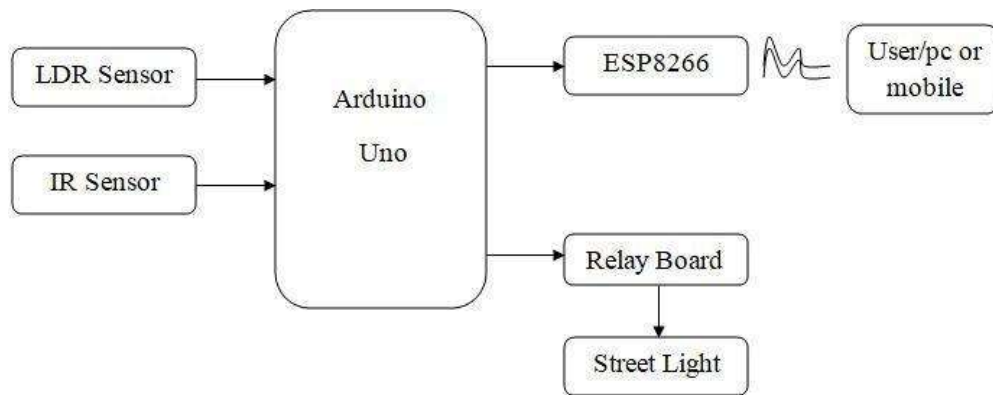


2.2 Smart Lighting

It is for roads, parks and buildings can help in saving energy. With smart lighting, city authorities can keep real-time tracking of lighting to ensure optimized illumination and deliver demand-based lighting in different zones. Smart lighting also helps in daylight harvesting and save energy by dimming out sectors with no occupancies for e.g. parking lots can be dimmed during work hours and when a car is entering, it will be detected and appropriate sectors can be illuminated, while others can be kept at diffused setting.

2.3 Smart Roads:

Equipped with sensors can provide information on driving condition, travel time estimating and alert in case of poor driving conditions, traffic condition and accidents. Such information can help in making the roads safer and help in reducing traffic jams. Information sensed from the roads can be communicated via Internet to cloud applications and social media and disseminated to the drivers who subscribe to such applications.

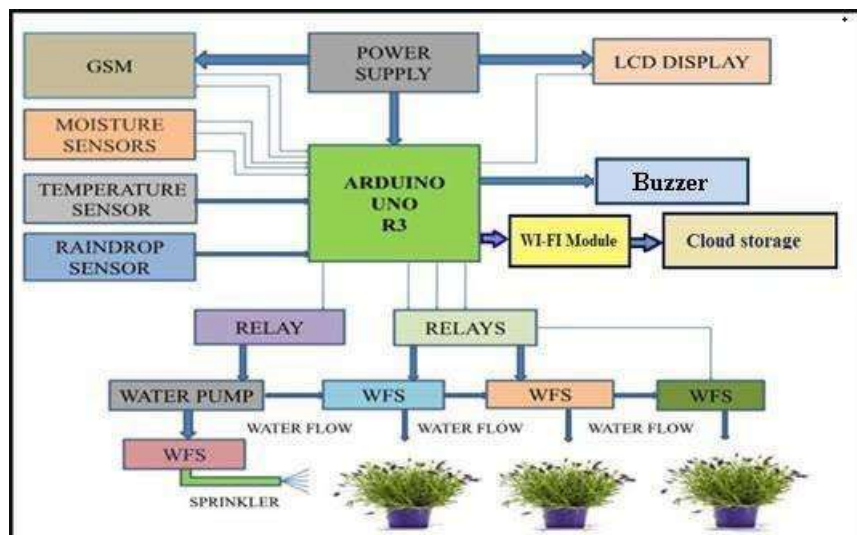


3. Structural Health Monitoring:

It uses a network of sensors to monitor the vibration levels in the structures such as bridges and buildings. The data collected from these sensors is analyzed to assess the health of the structures.

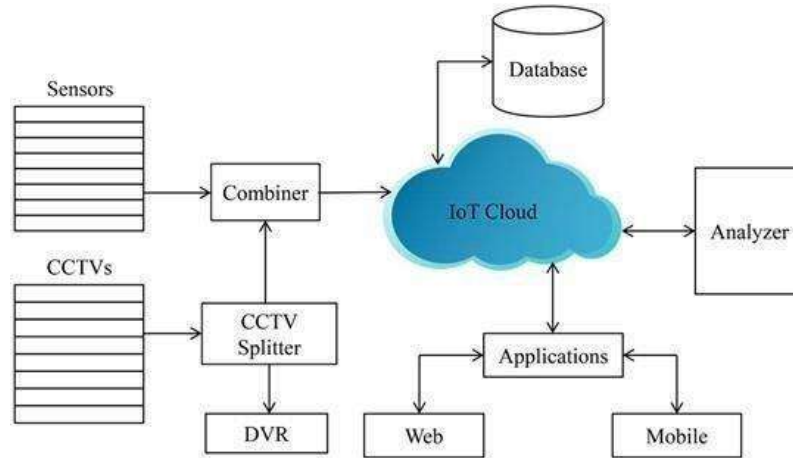
By analyzing the data, it is possible to detect cracks and mechanical breakdowns, locate the damages to a structure and also calculate the remaining life of the structure.

Using such systems, advance warnings can be given in the case of imminent failure of the structure.



4. Surveillance

The video feeds from surveillance cameras can be aggregated in cloud based scalable storage solution. The video feeds from surveillance cameras can be aggregated in cloud-based scalable storage solutions.



Cloud-based video analytics applications can be developed to search for patterns of specific events from the video feeds.



4.1 Emergency Response:

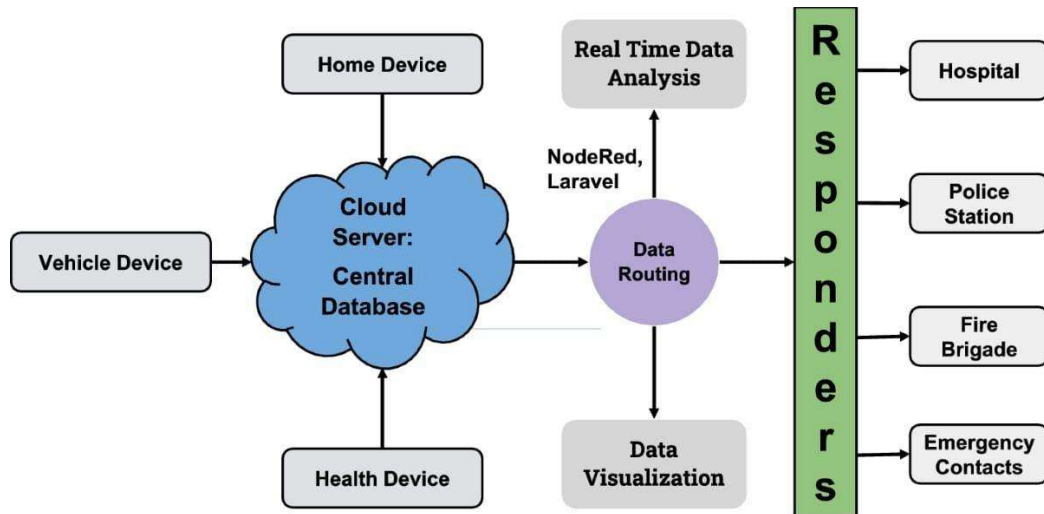
IoT systems for fire detection, gas and water leakage detection can help in generating alerts and minimizing their effects on the critical infrastructures. IoT systems can be used for monitoring the critical infrastructure cities such as buildings, gas, and water pipelines, public transport and power substations.

IoT systems for critical infrastructure monitoring enable aggregation and sharing of information collected from larger number of sensors.

Using cloud-based architectures, multi-modal information such as sensor data, audio, video feeds can be analyzed in near real-time to detect adverse events.

The alert can be in the form:

- Alerts sent to the public
- Re-rerouting of traffic
- Evacuations of the affected areas



5. ENVIRONMENT:

5.1 Weather Monitoring:

Systems collect data from a no. of sensors attached and send the data to cloud based applications and storage back ends. The data collected in cloud can then be analyzed and visualized by cloud-based applications. Weather alert can be sent to the subscribed users from such applications.

AirPi is a weather and air quality monitoring kit capable of recording and uploading information about temperature, humidity, air pressure, light levels, UV levels, carbon monoxide, nitrogen dioxide and smoke level to the Internet.

5.2 Air Pollution Monitoring:

System can monitor emission of harmful gases (CO₂, CO, NO, NO₂ etc.,) by factories and automobiles using gaseous and meteorological sensors. The collected data can be analyzed to make informed decisions on pollutions control approaches.

5.3 Noise Pollution Monitoring:

Due to growing urban development, noise levels in cities have increased and even become alarmingly high in some cities. IoT based noise pollution monitoring systems use a no. of noise monitoring systems that are deployed at different places in a city. The data on noise levels from the station is collected on servers or in the cloud. The collected data is then aggregated to generate noise maps. It uses a number of noise monitoring stations that are deployed at different places in a city.

The data on noise levels from the stations is collected on servers or in the cloud and then the collected data is aggregate to generate noise maps.

5.4 Forest Fire Detection:

Forest fire can cause damage to natural resources, property and human life. Early detection of forest fire can help in minimizing damage. IoT based forest fire detection system use a number of monitoring nodes deployed at different location in a forest.

Each monitoring node collects measurements on ambient condition including temperature, humidity, light levels, etc.

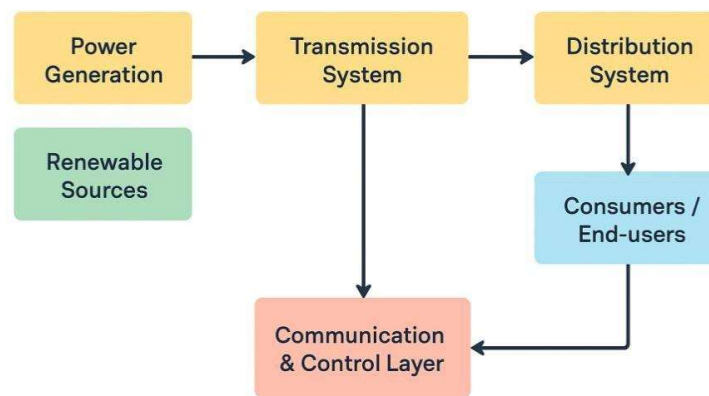
Early detection of forest fires can help in minimizing the damage.

5.5 River Flood Detection:

River floods can cause damage to natural and human resources and human life. Early warnings of floods can be given by monitoring the water level and flow rate. IoT based river flood monitoring system uses a no. of sensor nodes that monitor the water level and flow rate sensors. IoT based river flood monitoring system uses a number of sensor nodes that monitor the water level using ultrasonic sensors and flow rate using velocity sensors.

Data from these sensors is aggregated in a server or in the cloud, monitoring applications raise alerts when rapid increase in water level and flow rate is detected.

6. ENERGY:



6.1 Smart Grids:

It is a data communication network integrated with the electrical grids that collects and analyze data captured in near-real-time about power transmission, distribution and consumption. Smart grid technology provides predictive information and recommendations to utilities, their suppliers, and their customers on how best to manage power. By using IoT based sensing and measurement technologies, the health of equipment and integrity of the grid can be evaluated.

Smart grid technology provides predictive information and recommendation s to utilize, their suppliers, and their customers on how best to manage power.

Smart grid collects the data regarding:

- Electricity generation
- Electricity consumption
- Storage
- Distribution and equipment health data

By analyzing the data on power generation, transmission and consumption of smart grids can improve efficiency throughout the electric system.

Storage collection and analysis of smart grids data in the cloud can help in dynamic optimization of system operations, maintenance, and planning.

Cloud-based monitoring of smart grids data can improve energy usage levels via energy feedback to users coupled with real-time pricing information.

Condition monitoring data collected from power generation and transmission systems can help in detecting faults and predicting outage.

6.2 Renewable Energy Systems:

IoT based systems integrated with the transformers at the point of interconnection measure the electrical variables and how much power is fed into the grid. For wind energy systems, closed-loop controls can be used to regulate the voltage at point of interconnection which coordinate wind turbine outputs and provides power support. Due to the variability in the output from renewable energy sources (such as solar and wind), integrating them into the grid can cause grid stability and reliability problems.

- IoT based systems integrated with the transformer at the point of interconnection measure the electrical variables and how much power is fed into the grid.
- To ensure the grid stability, one solution is to simply cut off the overproductions.

6.3 Prognostics:

In systems such as power grids, real-time information is collected using specialized electrical sensors called Phasor Measurement Units (PMUs) at the substations. The information received from PMUs must be monitored in real-time for estimating the state of the system and for predicting failures. IoT based prognostic real-time health management systems can predict performance of machines of energy systems by analyzing the extent of deviation of a system from its normal operating profiles.

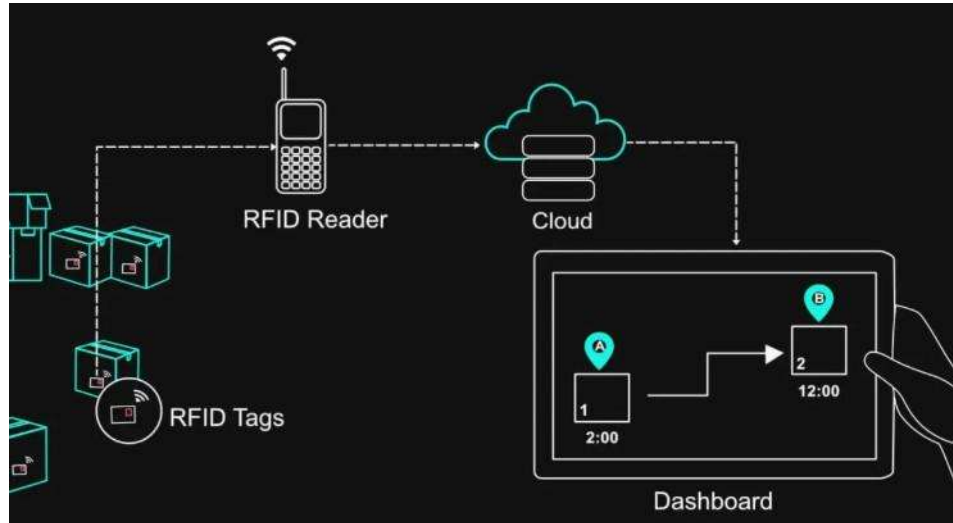
- In the system such as power grids, real time information is collected using specialized electrical sensors called Phasor Measurement Units (PMU)
- Analyzing massive amounts of maintenance data collected from sensors in energy systems and equipment can provide predictions for impending failures.
- OpenPDC is a set of applications for processing of streaming time-series data collected

from Phasor Measurements Units (PMUs) in real-time.

7. RETAIL

7.1 Inventory Management:

IoT systems enable remote monitoring of inventory using data collected by RFID readers.



IoT system using Radio Frequency Identification (RFID) tags can help inventory management and maintaining the right inventory levels.

- RFID tags attached to the products allow them to be tracked in the real-time so that the inventory levels can be determined accurately and products which are low on stock can be replenished.
- Tracking can be done using RFID readers attached to the retail store shelves or in the warehouse.

7.2 Smart Payments:

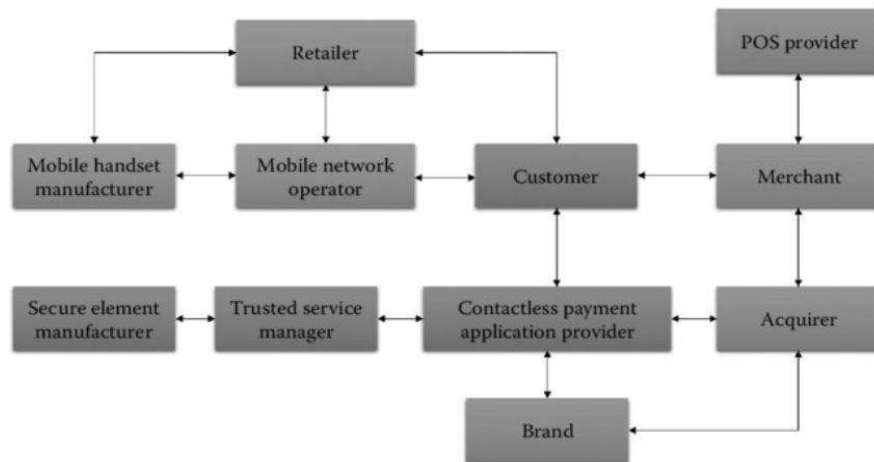
Solutions such as contact-less payments powered by technologies such as Near Field Communication (NFC) and Bluetooth.

- Smart payments solutions such as contact-less payments powered technologies such as Near field communication (NFC) and Bluetooth.
- NFC is a set of standards for smart-phones and other devices to communicate with each other by bringing them into proximity or by touching them.
- Customer can store the credit card information in their NFC-enabled smart-phones and make payments by bringing the smart-phone near the point-of-sale terminals.
- NFC maybe used in combination with Bluetooth, where NFC initiates initial pairing of devices to establish a Bluetooth connection while the actual data transfer takes place over Bluetooth.

7.3 Smart Vending Machines:

Sensors in a smart vending machine monitors its operations and send the data to cloud which can be used for predictive maintenance.

Smart vending machines connected to the Internet allow remote monitoring of inventory levels, elastic pricing of products, promotions, and contact-less payments using NFC.



Smart-phone applications that communicate with smart vending machines allow user preferences to be remembered and learned with time. E.g: when a user moves from one vending machine to the other and pair the smart-phone, the user preference and favourite product will be saved and then that data is used for predictive maintenance.

- Smart vending machines can communicated each other's, so if a product out of stock in a machine, the user can be routed to nearest machine
- For perishable items, the smart vending machines can reduce the price as the expiry date nears.

8. LOGISTICS

8.1 Route generation & scheduling:

IoT based system backed by cloud can provide first response to the route generation queries and can be scaled upto serve a large transportation network. Route generation and scheduling systems can generate end to end routes using combination of route patterns and transportation modes and feasible schedules based on the availability of vehicles IoT based system backed by cloud can provide first response to the route generation queries and can be scaled upto serve a large transportation network.

8.2 Fleet Tracking:

Use GPS to track locations of vehicles in real-time. Vehicle fleet tracking systems use GPS technology to track the locations of the vehicles in the real-time.

Cloud-based fleet tracking systems can be scaled up on demand to handle large number of vehicles,

- The vehicle locations and routers data can be aggregated and analyzed for detecting bottlenecks in the supply chain such as traffic congestions on routes, assignments and generation of alternative routes, and supply chain optimization.
- GPS to track locations of vehicles in real-time. Some of the main features that the modern Fleet Manager in the fleet tracking system are: route optimization, increased productivity, driver & vehicle safety and time management.

3 types of fleet tracking systems.

- i. Cellular
- ii. wireless passive
- iii. satellite.

8.3 Shipment Monitoring:

IoT based shipment monitoring systems use sensors such as temp, humidity, to monitor the conditions and send data to cloud, where it can be analyzed to detect food spoilage. Shipment monitoring solutions for transportation systems allow monitoring the conditions inside containers.

E.g : Containers carrying fresh food produce can be monitored to prevent spoilage of food. IoT based shipment monitoring systems use sensors such as temperature, pressure, humidity, for instance, to monitor the conditions inside the containers and send the data to the cloud, where it can be analyzed to detect food spoilage.

8.4 Remote Vehicle Diagnostics:

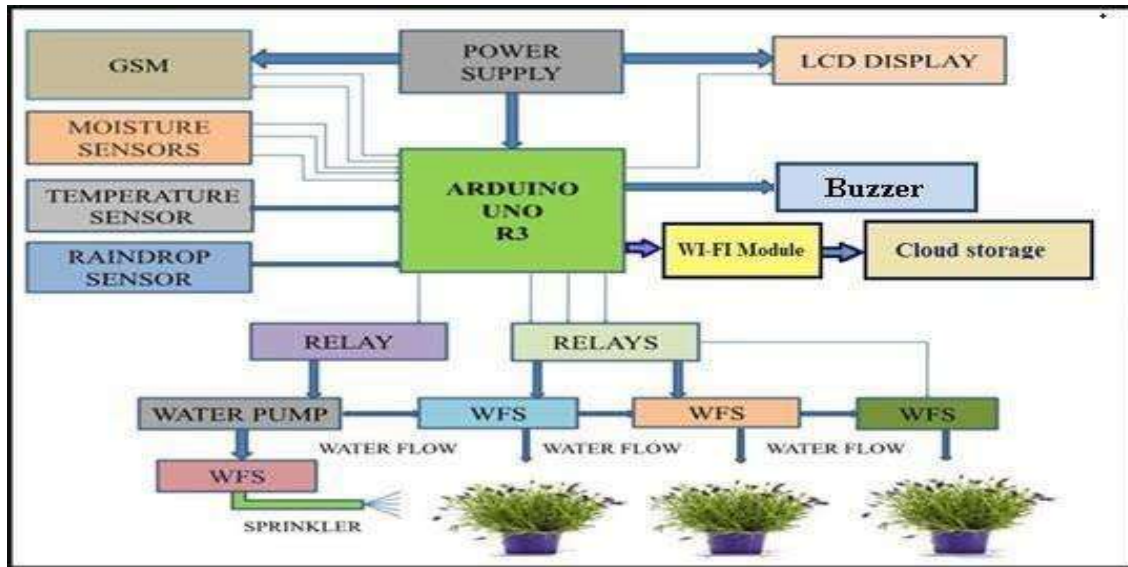
Systems use on-board IoT devices for collecting data on Vehicle operations (speed, RPM etc.) and status of various vehicle subsystems. It can detect faults in the vehicles or warn of impending faults.

- These diagnostic systems use on-board IoT devices for collecting data on vehicle operation such as speed, engine RPM, coolant temperature, fault code number and status of various vehicle sub-system.
- Modern commercial vehicles support on-board diagnostic (OBD) standard such as OBD-II

- OBD systems provide real-time data on the status of vehicle sub-systems and diagnostic trouble codes which allow rapidly identifying the faults in the vehicle.

IoT based vehicle diagnostic systems can send the vehicle data to centralized servers or the cloud where it can be analyzed to generate alerts and suggest remedial actions.

9. AGRICULTURE



9.1 Smart Irrigation

Smart irrigation system can improve crop yields while saving water.

- Smart irrigation systems use IoT devices with soil moisture sensors to determine the amount of moisture on the soil and release the flow of the water through the irrigation pipes only when the moisture levels go below a predefined threshold.
- It also collects moisture level measurements on the server on in the cloud where the collected data can be analyzed to plan watering schedules.
- Cultivar's Rain Cloud is a device for smart irrigation that uses water valves, soil sensors, and a WiFi enabled programmable computer. To determine moisture amount in soil.

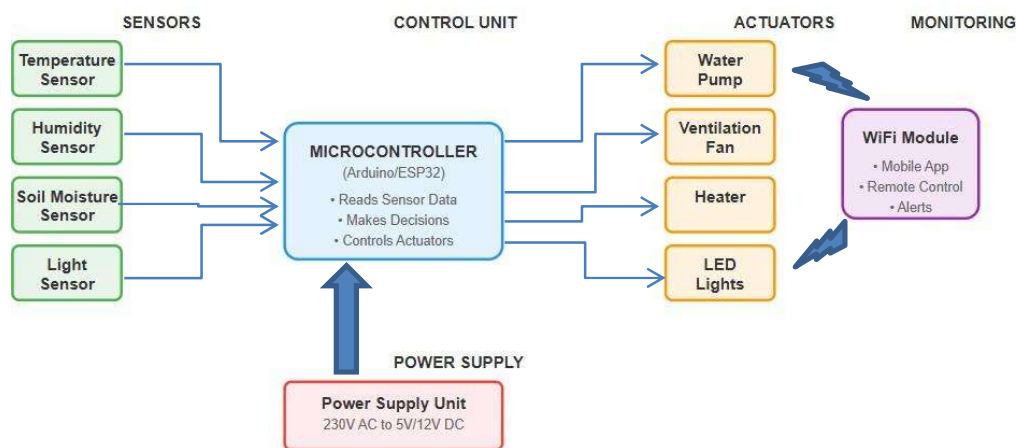
9.2 Green House Control: to improve productivity.

- Green Houses are structure with glass or plastic roofs that provide conducive environment for growth of plants.
- The climatological conditions inside a Green House can be monitored and controlled to provide the best conditions for growth of plants.
- The temperature, humidity, soil moisture, lightens, Carbon-di Oxide levels are

monitored using sensors.

- It controls temperature, humidity, soil, moisture, light, and carbon dioxide level that are monitored by sensors and climatological conditions that are controlled automatically using actuation devices.
- IoT systems play an importance role in green house control and help in improving productivity.
- The data collected from various sensors is stored on centralized servers or in the cloud where analysis is performed to optimize the control strategies and also correlate the productivity with different control strategies.

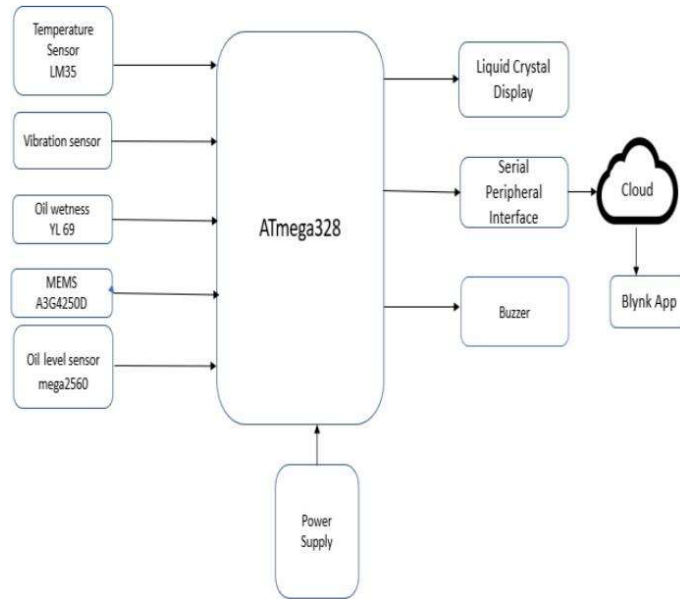
SIMPLE GREENHOUSE CONTROL SYSTEM



10. Industry

10.1 Machine Diagnosis and Prognosis

Machine prognosis refers to predicting the performance of a machine by analyzing the data on the current operating conditions and how much deviations exist from the normal operating conditions. Machine diagnosis refers to determining the cause of a machine fault. IoT plays a major role in both prognosis and diagnosis of industrial machines. Industrial machines have a large number of components that must function correctly for the machine to perform its operations. Sensors in machines can monitor the operating conditions such as (temperature and vibration levels). The sensor data measurements are done on timescales of few milliseconds to few seconds, which leads to generation of massive amount of data. IoT based systems integrated with cloud-based storage and analytics back-ends can help in storage, collection and analysis of such massive scale machine sensor data.



A number of methods have been proposed for reliability analysis and fault prediction in machines. Case-based reasoning (CBR) is a commonly used method that finds solutions to new problems based on past experience. This past experience is organized and represented as cases in a case-base. CBR is an effective technique for problem solving in the fields in which it is hard to establish a quantitative mathematical model, such as machine diagnosis and prognosis. Since for each machine, data from a very large number of sensors is collected, using such high dimensional data for creation of case library reduces the case retrieval efficiency. Therefore, data reduction and feature extraction methods are used to find the representative set of features which have the same classification ability as the complete features.

10.2 Indoor Air Quality Monitoring

Monitoring indoor air quality in factories is important for health and safety of the workers. Harmful and toxic gases such as carbon monoxide (CO), nitrogen monoxide (NO), Nitrogen Dioxide (NO₂), etc., can cause serious health problems. IoT based gas monitoring systems can help in monitoring the indoor air quality using various gas sensors. The indoor air quality can vary for different locations. Wireless sensor networks based IoT devices can identify the hazardous zones, so that corrective measures can be taken to ensure proper ventilation. A hybrid sensor system for indoor air quality monitoring is presented, which contains both stationary sensors (for accurate readings and calibration) and mobile sensors (for coverage). In addition, a wireless solution for indoor air quality monitoring is described, which measures environmental parameters such as temperature, humidity, gaseous pollutants, aerosols, and particulate matter to determine the indoor air quality.

11. Health & Lifestyle

11.1 Health & Fitness Monitoring

Wearable IoT devices that allow non-invasive and continuous monitoring of physiological parameters can help in continuous health and fitness monitoring. These wearable devices may be in various forms such as belts and wrist-bands. The wearable devices form a type of wireless sensor networks called body area networks in which the measurements from a number of wearable devices are continuously sent to a master node (such as a smart-phone) which then sends the data to a server or a cloud-based back-end for analysis and archiving. Health-care providers can analyze the collected health-care data to determine any health conditions or anomalies. Commonly used body sensors include: body temperature, heart rate, pulse oximeter oxygen saturation (SpO₂), blood pressure, electrocardiogram (ECG), movement (with accelerometers), and electroencephalogram (EEG). A ubiquitous mobility approach for body sensor networks in health care has been proposed. A wearable ubiquitous health-care monitoring system is presented that uses integrated electrocardiogram (ECG), accelerometer, and oxygen saturation (SpO₂) sensors. The Fitbit wristband is a wearable device that tracks steps, distance, and calories burned during the day, as well as sleep quality at night.

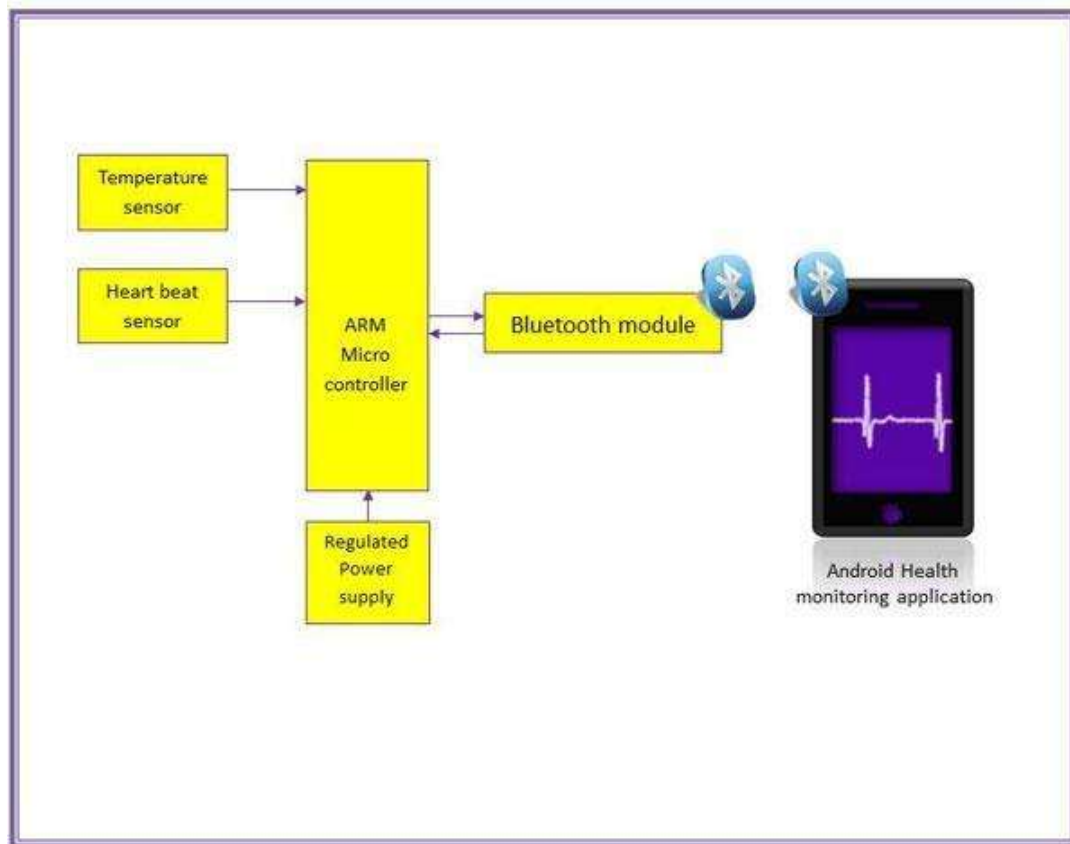


Figure. Applications of IoT for health

11.2 Wearable Electronics

Wearable electronics such as wearable gadgets (smart watches, smart glasses, wristbands, etc.) and fashion electronics (with electronics integrated into clothing and accessories, e.g., Google Glass or Moto 360 smart watch) provide various functions and services to assist us in our daily activities and making us lead healthy lifestyles. Smart watches that run mobile operating systems (such as Android Wear) can be used as portable communication and time keeping. With smart watches, the users can record their daily activities and exercise, make calls with or without paired mobile phones, check calendar schedules, and download mobile applications. Smart glasses allow users to take photos and videos, check e-mails, and access web applications by projecting the images onto the glasses lens. Smart shoes monitor the walking or running speed and jump with the help of built-in sensors and can be integrated with smart-phones to visualize the data. Smart wristbands can track the daily exercise and calories burnt.